

MD. ALIF HOSSEN

ENTRY-LEVEL MACHINE LEARNING ENGINEER | APPLIED AI DEVELOPER

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PROFESSIONAL SUMMARY

Computer Science and Engineering graduate focused on machine learning, applied AI, and data-driven applications. Builds Python-based end-to-end systems spanning data pipelines, model validation, RAG, computer vision, REST APIs, testing, monitoring, and Streamlit deployment.

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, Scikit-learn, feature engineering, data analysis
ML & Applied AI: LightGBM, XGBoost, time-series forecasting, SHAP, RAG, BM25, FAISS, NLP
Computer Vision: YOLOv8, OpenCV, EasyOCR, object detection, tracking, OCR
Applications & Delivery: FastAPI, REST, Pydantic, Streamlit, PostgreSQL, SQLite, Docker, GitHub Actions, pytest
Analytics & Visualization: Matplotlib, Seaborn, Plotly, Power BI, Microsoft Excel, Jupyter

SELECTED PROJECTS

- SEC Filings RAG Intelligence** | [Live Demo](#) | [GitHub](#)
 - Engineered citation-grounded research over SEC 10-K/10-Q filings using section-aware processing, hybrid BM25 + FAISS retrieval, reciprocal-rank fusion, and cross-encoder reranking.
 - Integrated SEC XBRL Company Facts and deterministic financial calculations; delivered the workflow through FastAPI, Streamlit, tests, CI, and containerization.**Stack:** Python, BM25, FAISS, SentenceTransformers, FastAPI, Streamlit, SQLite, Docker
- Real-Time Transaction Fraud Risk & Monitoring Platform** | [Live Demo](#) | [GitHub](#)
 - Developed a fraud-risk pipeline with temporal data separation, probability calibration, validation-based model selection, and cost-sensitive decision thresholds.
 - Implemented SHAP reason codes, API inference, interactive scoring, monitoring modules, automated tests, and optional PostgreSQL persistence.**Stack:** Python, Scikit-learn, LightGBM, XGBoost, SHAP, FastAPI, Streamlit, PostgreSQL
- Retail Demand Forecasting & Inventory Optimization** | [Live Demo](#) | [GitHub](#)
 - Built an M5 forecasting workflow with leakage-safe lag/rolling features, Seasonal Naive and LightGBM models, chronological validation, and rolling-origin backtesting.
 - Translated forecast uncertainty into safety stock, reorder points, order quantities, and stockout/overstock risk through a FastAPI-backed Streamlit application.**Stack:** Python, Pandas, LightGBM, Time Series, FastAPI, Streamlit, Docker
- AI-Based Real-Time Traffic Violation Detection System** | [Live Demo](#) | [GitHub](#)
 - Built an academic road-video screening prototype using YOLOv8 vehicle detection and tracking, custom plate-region detection, and EasyOCR transcription.
 - Delivered annotated video, vehicle crops, summary counts, and CSV exports through Streamlit, with heuristic events explicitly requiring human review.**Stack:** Python, YOLOv8, OpenCV, EasyOCR, NumPy, Pandas, Streamlit

EDUCATION

QUALIFICATION	INSTITUTION	RESULT	YEAR
B.Sc. in CSE	Daffodil International University	CGPA 3.33 / 4.00	July 2026
HSC (Science)	Shahid Mamun Mahmud Police Lines School & College, Rajshahi	GPA 5.00 / 5.00	2021
SSC (Science)	Goal Manda High School	GPA 5.00 / 5.00	2019

CERTIFICATIONS

- Postman API Fundamentals Student Expert | Postman | Nov 2025
- Basics of Machine Learning Algorithms | UniAthena | Oct 2025
- Basics of Data Science | UniAthena | Oct 2025
- ChatGPT Basics, Usage and Prospects (Distinction) | GoEdu | Nov 2024